Draft, not submitted. Publication and current-version dates are assigned by IEEE.

Digital Object Identifier To be assigned by IEEE

Evidence-Gated Crop-Loss Claim Verification and Profile-Aware Advisory Chat for Smallholder Farmers: Design, Conformance Evaluation and Live Feasibility Study

AUTHOR NAMES WITHHELD IN THIS DRAFT[1]
[1]Affiliations withheld in this draft

Corresponding author: to be named before submission.

Support and funding information will be added by the authors before submission.

ABSTRACT Smallholder crop-loss claims are hard to verify, because no single evidence source establishes who may claim or whether a loss occurred. We describe Vivasayi AI, a farmer-facing system for Tamil Nadu, India, that pairs a profile-aware bilingual advisory chat with claim verification in which every evidence source is reduced to an explicit per-component outcome and a deterministic, versioned rule engine, never a language model, decides. We compare three policies: the historical photograph-only policy; a safe-AI policy, the default as of the change of 16 October 2026, under which neither a model reading alone nor a technical failure can cause an adverse decision; and a required-component gate that makes approval depend on claimant-rights evidence but is not the default, because no authoritative claimant-rights source is available. On a synthetic factorial whose ground truth is the design rule, the photograph-only and safe-AI policies each approved 10 of 190 forbidden scenarios and the gate none; only the photograph-only policy rejected claims on the AI reading alone. In a live feasibility study the production Sentinel-2 step ran on 12 parcel-sized areas: near three documented 2023–2024 flood and cyclone events it reported no supporting loss in any of 9 areas, and on a harvest control it falsely reported loss in 1 of 3. Live retrieval found a relevant chunk for only 2 of 13 answerable questions. Live language-model evaluations were blocked by a free-tier quota. *No result is a measurement on real claims; they show rule enforcement and feasibility, not accuracy.*

INDEX TERMS Agricultural advisory, auditability, claim verification, crop insurance, evidence fusion, NDVI, remote sensing, retrieval-augmented generation, Sentinel-2, Tamil.

I. INTRODUCTION

INDEMNITY and index crop insurance depend on knowing whether a loss occurred, on which land, and who may claim it. Remote sensing is used to design area indices, to monitor parcels and to smart-sample crop-cutting experiments [1]–[3], and India's YES-TECH programme blends modelled and crop-cutting yields at insurance-unit level [4]. Basis risk remains, and resolution and timing affect it [5]–[7]. Picture-based insurance has been proposed as a farmer-level complement [8]. Trust is a barrier to insurance take-up in India [9], and duplicate insurance of the same land has been reported in the press [10]. At the same time, language and vision-language models can read documents and photographs [11], [12], but hallucination is a documented risk [13], and for high-stakes decisions interpretable rules and audit trails are preferred to opaque models [14], [15].

This paper describes a system built on that premise: models produce *evidence*; a deterministic engine produces the *decision*; and every decision is tied to a versioned policy and a snapshot of the evidence it used. Our contributions are:

1) A verification design in which each evidence source maps to an explicit per-component outcome, provider failure and missing evidence are kept distinct from negative findings, and versioned policies change how a candidate outcome becomes a decision without changing the frozen rules: a safe-AI policy under which neither a model reading alone nor a technical failure causes an adverse decision, and a required-component gate that can only make an approval more cautious, holding the claim rather than deciding when evidence is pending.
2) An open implementation: a farmer-facing web ap-

plication and backend with context-sufficiency logic for bilingual chat, parcel-geometry validation, cross-farmer overlap as a review signal, Tamil/Latin name matching, and a pre/post-event Sentinel-2 NDVI step on the claim polygon with scene-classification masking.

3) A conformance evaluation on synthetic benchmarks and a live feasibility study (satellite, retrieval, end-to-end farmer journeys), with every post-hoc change disclosed, and a protocol for the real-data experiments that remain.

We do not claim novelty of the individual techniques, which have precedent, nor any accuracy on real claims.

II. RELATED WORK AND RESEARCH GAP

Satellite crop-damage assessment. Sentinel-2 [16] and NDVI [17] are standard for vegetation monitoring. Pre/post-event NDVI differencing has mapped flood damage to crops, with the authors noting the lack of direct ground validation [18]; Sentinel-2 has been used for hail [19] and flood damage with in-situ data [20], and for land-use anomaly checks in EU subsidy monitoring [21]. Closest to our decision logic, a rule-based decision matrix fusing Sentinel-1 and Sentinel-2 evidence mapped jute lodging after Cyclone Amphan in India [22]. Cloud and shadow masking needs validation [23]–[25], and Tamil Nadu has a public field-level crop dataset [26]. Satellite yield estimation can track ground measures on smallholder plots [27], [28].

Agricultural assistants and grounding. Retrieval-augmented generation [29], [30] has been applied to agriculture with domain constraints [31], compared with fine-tuning [32], and deployed as personalised farmer chat with LLM-judged faithfulness [33]. Agricultural question-answering benchmarks and systems include AgXQA [34] and Kisan-QRS [35]; AgriLLM studied Kisan Call Centre queries from Tamil Nadu [36]. Whether retrieved context is *sufficient* to answer has been proposed as a lens on retrieval failures [37]; automated RAG metrics exist [38]. Voice advisory in India has been evaluated [39], and vision-language models remain weak on fine-grained agricultural recognition [40].

Gap. In a search of 45 sources verified from abstracts and bibliographic records plus 3 web documents, extended by 26 sources in a second pass (16 October 2026; the bibliography lists 74 entries, and a source ledger records how much of each was read), we found rule-based fusion of satellite evidence for damage mapping [22] and farmer-level photograph insurance [8], but no work that gates individual crop-loss claims on claimant-rights, geometry, photograph, document, satellite and weather components with explicit failure semantics, treats cross-farmer polygon overlap as a review signal, or matches Tamil/Latin names on land documents; nor a public labelled dataset of Indian claim outcomes. Absence in a narrow search is weak evidence, so we make no priority claim.

III. SYSTEM DESIGN

The backend is a Node.js service on AWS Lambda with MongoDB; the farmer-facing client is a React web application in English and Tamil. A farmer registers parcels as polygons; a claim is made against a registered parcel with a drawn affected area and a declared relationship to the land. The server computes area, validates geometry, and never accepts area, state or decision from the client.

A. EVIDENCE COMPONENTS AND OUTCOMES

Ten components describe a claim: parcel identity, claimant rights, land documents, geometry, crop, satellite, photographs, weather, overlap and manual review. Each resolves to one of seven outcomes. A provider that errors or is disabled yields *unavailable*; missing evidence yields *unverified*; *failed* requires a positive finding that contradicts the claim. This vocabulary is the same under every policy; how a technical failure reaches the decision is policy-specific (Section III-B). Evidence derived by a model (photographs, document reading) is never authoritative, and a model reading alone can never verify claimant rights. Satellite and weather are supporting evidence and are never *failed*. Evidence is append-only: a re-run with new inputs adds a row, and the newest row per component is current.

B. DETERMINISTIC DECISION AND THE THREE POLICIES

A frozen rule engine applies, in order: claim window, event type, area allowance, containment in the parcel, duplicate or partial overlap with the farmer's own claims, remaining eligible area, and the photograph assessment. A versioned policy then decides how that candidate outcome becomes a decision. Each decision stores the policy identifier and a hash of its content, the component outcomes and references to the evidence rows used, so an earlier decision replays under its own policy. The repository defines three policies:

- `v1-legacy`, the historical behaviour and the policy of the live study. Component outcomes are recorded, but none blocks a decision. A confident AI "no damage" reading alone rejects the claim. A provider error or timeout leaves the claim undecided and retryable, but an unreadable model answer is treated as an uncertain photograph, and the farmer is asked for more evidence.
- `v1.1-safe-ai`, the default as of the change of 16 October 2026: the legacy rules with four changes. (i) An AI-only "no damage" reading holds the claim for human review and opens one review case instead of rejecting it; a rejection by another rule (claim window, event type, area outside the parcel) stands. (ii) A technical failure, including an unreadable model answer, makes the photograph check *unavailable*: the queue retries it, and the engine holds the claim instead of rejecting it or asking for more evidence. (iii) The claim window is measured at the claim's first submission, not at decision time. (iv) The decision never waits for satellite evidence.
- `v2-components`, the required-component gate. Parcel identity, geometry, photographs, overlap and

claimant rights are required. An approval whose required component is unverified or failed becomes a request for more evidence; one whose required component is pending, unavailable, inconclusive or needs review is *held*; and a rejection that coincides with a confident AI "no damage" reading is held for review. The gate was evaluated but is not the default. A model reading of a document can never verify claimant rights, and no authoritative claimant-rights provider (an authorised land-record service) is available to us, so under the gate no claim could be approved before a reviewer had accepted a land document.

A held claim stays submitted and withdrawable, and no decision is recorded; an administrator may decide it through an audited override. Neither the gate nor a hold ever produces a rejection; rejection stays with the frozen rules and that override. Only under `v1-legacy` can a model reading alone reject a claim. Under `v1-legacy` and `v2-components` an unreadable model answer returns the claim to the farmer for more evidence; under `v1.1-safe-ai` no technical failure does.

Timing. Slow evidence steps run from a database-backed job queue with leases, so a Lambda that is frozen mid-step cannot lose work. Satellite runs as its own step, independently and including later reschedules while its post-event window is open, and never blocks or changes a decision under any policy: the engine waits for the other queued checks but not for satellite, whose result only adds a supporting evidence row. (In the code of the live study a decision could wait for the post-event window to close.) Under `v1.1-safe-ai` the claim window is measured at the claim's first submission, which is recorded once and kept on resubmission, so time spent waiting for checks, a resubmission or a review cannot turn into a late-claim rejection. `v1-legacy` and `v2-components` still measure it at decision time, so their stored decisions replay unchanged.

Automatic processing. After the study, submission was changed to record the claim's plan of checks durably and idempotently within the request; a worker (a local process and a scheduled Lambda handler) runs due jobs without a browser open, renewing leases and sweeping stuck or lost work; and a farmer-facing tracker shows per-stage progress built only from persisted state, with a satellite status that is never shown as passed unless an analysis supported the loss. These are implemented and covered by unit and integration tests. None is deployed, and the scheduled worker's trigger is disabled.

C. GEOMETRY AND OVERLAP

Polygons are validated (closed ring, finite in-range coordinates, no self-intersection, non-zero area). Containment uses a tolerance. Overlap with another farmer's claim is found through a bounding-box pre-filter and an exact intersection, and is a review signal only, since shared boundaries, joint holdings and cadastral error are legitimate causes.

D. DOCUMENTS AND CLAIMANT RELATIONSHIP

A vision-language model reads a land document into structured fields; the document is untrusted data, and instruction-like text inside it is itself a finding. Holder names are compared with the claimant across Tamil and Latin scripts by a consonant-skeleton matcher. Each declared relationship (owner, joint owner, tenant, authorised cultivator, heir) has its own expected document set. No authorised Tamil Nadu land-record API is available to us; the official-record adapter is provider-neutral and reports *unavailable*.

E. SATELLITE EVIDENCE

For each claim the system requests per-pixel NDVI statistics over the claim polygon from the Copernicus Data Space Statistical API [41] for a pre-event window (event -40 to -5 days) and a post-event window ($+1$ to $+20$ days, capped at the present), masking pixels the Sentinel-2 L2A scene classification marks as no-data, saturated, cloud shadow, cloud or cirrus. Water is deliberately not masked, so flooding stays observable, which is also a confound for paddy. A drop in mean NDVI of at least 0.15, with a valid-pixel fraction of at least 0.6 in both windows, is reported as *supporting loss*; otherwise *no change* or *inconclusive*. The thresholds are provisional and uncalibrated. A change in a vegetation index does not identify a cause. Historical rainfall [42] is recorded as supporting evidence only. Neither satellite nor weather evidence blocks or changes a decision (Section III-B).

F. PROFILE-AWARE ADVISORY CHAT

For each message the assistant classifies intent, loads only the context that intent needs from the farmer's own records (optionally a selected parcel), tags each item with its provenance (user-provided, verified, third-party, inferred, missing, stale), and asks a follow-up only when a required slot cannot be resolved, about exactly one thing. Retrieved knowledge is fenced and labelled as data, and its source files are returned to the client. Replies follow the requested or detected language.

IV. EVALUATION DESIGN

Conformance benchmarks. Deterministic, seeded and reproducible from the repository (`npm run eval`). Ground truth is analytic (geometry), the design rule (fusion), or a written specification (chat); the name set is author-labelled. No network or real data are used. The `v1.1-safe-ai` fusion row was computed after the other results, on the same factorial, and written to its own dated result file; the earlier rows still reproduce exactly.

Live feasibility study (16 October 2026). (i) *Satellite:* the production adapter, including the satellite changes made during the study and disclosed in Section VI, on 100 m squares near three documented events, south Tamil Nadu floods of 17–18 December 2023, Cyclone Michaung of 4 December 2023 and Cyclone Fengal of 30 November–1 December 2024, located from humanitarian situation reports, and on a Thanjavur control dated 15 February 2024, within the samba harvest season, with three squares each. Land cover,

TABLE 1. Geometry checks against analytic ground truth (synthetic rectangles placed across Tamil Nadu).

Check	*n*	Precision	Recall
Outside-parcel detection	300	1	1
Cross-farmer overlap flag	300	1	1
Invalid-geometry detection	190	1	1

Median relative area difference from an ellipsoidal reference: 0.63% (systematic).

TABLE 2. Follow-up question decisions, scored against a written specification (55 synthetic cases: 44 where no question is needed, 11 where one is).

	Previous rule	Context logic
Unnecessary questions asked	81.8% (36/44)	0.0% (0/44)
Needed questions missed	27.3% (3/11)	0.0% (0/11)
Crop assumed when ambiguous	100.0% (6/6)	0.0% (0/6)

Intent classification accuracy 100.0% (29/29). Context assembly latency p50/p95 0.37/0.75 ms with weather stubbed.

crop and actual damage at each square were not verified, so these are case studies, not labelled data. (ii) *Retrieval:* 22 queries (13 answerable from the deployed corpus, 9 advisory) against the production collection, read-only, with relevance by a mechanical rule fixed before the run. (iii) *Acceptance:* the application's farmer journey (profile, parcel, chat, claim, photographs, land document, submission, verification, cross-user access) run over HTTP against the real application on a local database with mock storage, mock image AI and mock land records, under `v1-legacy` and `v2-components` (the study predates `v1.1-safe-ai`). (iv) Live chat and synthetic land-document extraction were prepared but could not run (Section VII).

V. RESULTS

Geometry (Table 1). Containment, overlap flagging and invalid-geometry detection agree with analytic geometry. The harness exposed two issues: the shared validator accepted out-of-range coordinates (now rejected), and the server's area is about 0.6% larger than an ellipsoidal reference at Tamil Nadu latitudes because the formula uses the equatorial radius; we report rather than change it, so stored areas stay comparable.

Evidence fusion (Table 3). The legacy photograph-and-geometry policy approves claims with missing, pending or inconsistent ownership evidence and with overlap on another farmer's claim; the gate does not. `v1.1-safe-ai` does not reduce invalid approvals relative to `v1-legacy`: it approves the same 10 of 190 forbidden scenarios, so it guards against adverse decisions, not against wrongful approval. What it changes is the AI-alone rejection: `v1-legacy` rejected 2 of the 2 scenarios in which only the AI reading stood against the claim, `v1.1-safe-ai` rejected 0, and the only decisions that differ between the two policies are 12 rejections produced by the AI rule, which become holds for review. Satellite and weather outcomes never changed a decision in 1,536 comparisons per policy under any of the three policies (under `v1.1-safe-ai` they did not change a hold either), by design. Because the label is the design rule, this shows rule enforcement, not accuracy.

TABLE 3. Evidence fusion over a 192-scenario factorial: invalid approvals, claims rejected on the AI reading alone, and claims held for review.

Policy	Invalid approvals	Rejected on AI alone	Held
photo-only (v1-legacy)	10/190	2/2	0
full fusion (v2-components)	0/190	0/2	40
v2 without claimant_rights	4/190	0/2	42
v2 without overlap	2/190	0/2	38
v2 without photos	0/190	0/2	40
v2 without geometry	0/190	0/2	40
photo-only, safe AI (v1.1-safe-ai)	10/190	0/2	12

The v1.1-safe-ai row was computed on 2026-10-16 on the same factorial (fusion_v1.1_2026-10-16.json); the other rows are the published results, which still reproduce exactly.

TABLE 4. Simulated behaviour of the provisional NDVI-change rule (no real imagery; 400 trials per setting).

Setting	Outcome rate
No change, clear sky: flagged as supporting loss	0.0%
No change, cloudy: flagged as supporting loss	0.0%
NDVI drop 0.15, clear sky: detected	49.8%
NDVI drop 0.30, clear sky: detected	99.8%
Drop 0.30, 70% of pixels masked: inconclusive	100.0%

Chat context (Table 2). Against the previous rule, which asked a follow-up whenever a message ended with a question mark, the context logic asked no unnecessary questions and missed none on the specification. The specification and logic share an author, and the baseline is a reimplementation.

Satellite, simulated (Table 4) and live (Table 5). All 12 live runs completed in a median of 2.5 s (maximum 11.8 s). Near the three events the rule reported supporting loss in none of 9 squares: 5 showed no change, 1 had too few valid post-event pixels, and 3 returned no valid statistics for the post-event window. Three of the analysed event squares had a pre-event NDVI between -0.04 and 0.10, consistent with bare, flooded or recently transplanted fields, so a fixed window cannot observe a loss there and NDVI rose afterwards. On the harvest control one square reported supporting loss ($\Delta\text{NDVI} = -0.25$, from 0.74 to 0.49 across the harvest), which is a false support produced by normal crop phenology. These observations support the design choice that satellite evidence is optional and never decisive, and they show that a fixed window needs crop-calendar context before thresholds can be calibrated.

Retrieval (Table 6). The deployed corpus (450 chunks from 60 files) consists mostly of district statistics tables and a geography file; it includes a rice pest-prediction table (13 chunks) whose content we did not inspect, and no retrieved chunk could ground an advisory question (0 of 27 chunks). For questions the tables can answer, the right file was retrieved for 10 of 13, but in 9 of 11 such hits the chunk was a header without data rows; a relevant chunk was retrieved for only 2 of 13. The cause is in ingestion: rows are chunked without the file name or crop, so a district's row is rarely retrieved with its table. Source attribution therefore matters: a farmer and an auditor can see that an answer was not

TABLE 5. Live Sentinel-2 L2A runs of the production adapter on 100 m squares near documented events and a harvest control (run 2026-10-16). Land cover and crop at each square were not verified; the event rows are case studies, not labelled damage.

Event (date)	Squares	Supports loss	No change	Inconclusive	ΔNDVI
south_tn_floods_2023 (2023-12-18)	3	0	2	1	0.11, -0.02
cyclone_michaung_2023 (2023-12-04)	3	0	2	1	0.46, 0.31, 0.56
cyclone_fengal_2024 (2024-12-01)	3	0	1	2	0.03
thanjavur_harvest_control_2024 (2024-02-15)	3	1	1	1	-0.25, -0.07, 0.21

TABLE 6. Live retrieval on the production collection (farming-documents, embed-english-v3.0, top-3). Relevance by a mechanical author-written rule, not human judgement.

Measure	Rate (k/n)
Answerable: any relevant chunk	15.4% (2/13)
Answerable: top-1 relevant	7.7% (1/13)
Answerable: expected file retrieved	76.9% (10/13)
Expected-file hits that are header-only	81.8% (9/11)
Advisory: answer-bearing chunks	0.0% (0/27)

grounded.

Acceptance. Under `v1-legacy` and `v2-components` every journey step except live chat passed (24 of 26 under the legacy policy, 25 of 27 under the gated policy), including idempotent replay, cross-user access returning ‘‘not found’’, and a client-supplied verification result being ignored. Under the legacy policy the claim was approved on photographs and geometry with claimant rights unverified; under the gated policy it was returned for more evidence, and after a land record was attached and the claim resubmitted it was held with claimant rights pending review, the intended behaviour. No acceptance run under `v1.1-safe-ai` has been recorded.

Name matching. On 60 author-labelled pairs the matcher scored 55 of 60 before a rule change and 60 of 60 after; the change followed those errors on the same set, so the second figure is optimistic.

VI. DISCLOSURES: DEFECTS FOUND AND POST-HOC CHANGES

The live study found defects that no synthetic test had exposed. We list them because they bear on how far the conformance results can be trusted.

- *Response parsing.* The production NDVI parser read the provider’s statistics at the wrong nesting level, so every live analysis in our first runs returned ‘‘no valid statistics’’ (those runs were not saved). It was fixed, and a recorded live response is now a regression fixture.
- *Scene cloud filter (post-hoc).* A 30% scene-level cloud limit rejected all monsoon imagery for the event windows in our first live runs (not saved; recorded only in a code comment). We raised the scene limit used for parcel statistics to 90% after observing this, relying on per-pixel masking for the parcel itself. The NDVI thresholds were not changed.
- *Prototype request.* The earlier proof-of-concept request was rejected by the live service (HTTP 400, missing data-mask output); it had never run live. Production no longer uses it.
- *Quota masked as timeout.* A model-provider quota refusal was retried with back-off until the 25 s timeout fired, so farmers would have seen ‘‘did not respond in time’’. Retries are now capped and the refusal is reported as a temporary unavailability.
- *Stale decision on hold.* A claim held after a resubmission still reported the previous round’s decision. The hold now clears it; the earlier decision remains on the audit trail.
- *Write budget.* A realistic single-claim journey exceeds the per-user limit of five claim writes per hour in force during the study; that value is kept in a pinned record of the study-time code defaults. After the study the owner raised the defaults to fifteen claim writes and 60 evidence requests per user per hour (16 October 2026); this change is not deployed, and no study result depends on it.

VII. BLOCKED EVALUATIONS

The configured language-model key was on the provider’s free tier, and its daily request quota for the model used was exhausted during the study (the quota was diagnosed from the provider’s error during the study and is not recorded in a saved result). The live chat comparison (context-aware versus generic, 46 items) and synthetic land-document extraction (12 rendered, watermarked documents, 3 with injected instructions) are implemented and ready to run but produced no results, and we report none. Latency, token and cost figures for chat are therefore absent.

VIII. DISCUSSION

The structural finding is that separating evidence from decision, and failure from negative finding, makes behaviour under outages and missing data explicit and testable, and lets a stricter or safer policy be introduced without changing the frozen rules or the replay of earlier decisions. The safe-AI policy shows the limit of that separation as well as its use: it removes adverse decisions caused by a model reading or an outage, but it does not change which claims are approved. The live study adds two empirical cautions that the synthetic study could not: a fixed-window NDVI change rule is confounded by crop phenology and sparse post-event observation, and retrieval quality depends on corpus content and chunking more than on the retriever. Both favour the conservative roles the design assigns to these sources.

IX. LIMITATIONS AND THREATS TO VALIDITY

- No real claims, farmers or documents were used. No accuracy, fraud-detection or loss-estimation figure is claimed.
- Live satellite cases are 12 squares chosen near documented events without verifying land cover, crop or damage; they are not labelled data, and one parameter was changed after observation.
- Specifications and relevance rules are author-defined; the name matcher was tuned on its own evaluation set.
- `v1.1-safe-ai` is evaluated only on the synthetic factorial and by automated tests; no live or acceptance run used it, and technical failures are not a factor of the factorial.
- Live language-model behaviour is unmeasured. The deployed knowledge base holds almost no advisory text (one rice pest-prediction table, not inspected).
- No authorised land-record API is available; the required-document policy per relationship is unreviewed by any legal or insurer authority.

X. ETHICAL, LEGAL AND PRIVACY CONSIDERATIONS

No real farmer data were used; the synthetic documents are watermarked and use fictional names. Real use needs informed consent, secure handling of documents, permission for any official record and human review of adverse and uncertain outcomes; under `v1.1-safe-ai` and `v2-components` no model reading alone rejects a claim, while under `v1-legacy` one can. The statement in the repository has not been reviewed by a lawyer or ethics board.

XI. CONCLUSION AND FUTURE WORK

We described, implemented and tested an evidence-gated claim-verification design with versioned policies, including a safe-AI default, and a profile-aware advisory chat, and ran the satellite, retrieval and farmer-journey paths live. The results show that the components enforce their specifications and that the live paths work; they do not show real-world accuracy. Next steps are: live chat answers rated by agronomists in English and Tamil; labelled document extraction; satellite validation against independently labelled parcels with crop-calendar-aware windows and Sentinel-1 for cloudy periods; re-chunking the knowledge base with advisory sources; and evaluation on authorised cadastral data.

REFERENCES

[1] S. A. Okeyo and G. C. Mulaku, "Crop insurance based on satellite remote sensing: Case of migori county, kenya," *Advances in Remote Sensing*, vol. 09, no. 01, p. 1–11, 2020. [Online]. Available: http://dx.doi.org/10.4236/ars.2020.91001

[2] V. Sitokonstantinou, I. Papoutsis, C. Kontoes, A. Lafarga Arnal, A. P. Armesto Andrés, and J. A. Garraza Zurbano, "Scalable parcel-based crop identification scheme using sentinel-2 data time-series for the monitoring of the common agricultural policy," *Remote Sensing*, vol. 10, no. 6, p. 911, Jun. 2018. [Online]. Available: http://dx.doi.org/10.3390/rs10060911

[3] C. S. Murthy, R. Das, K. K. Choudhary, V. Pandey, P. Srikanth, and P. V. N. Rao, "Smart sampling for crop yield estimation in the insurance units of PMFBY: Technique development and implementation in Odisha state," https://pmfby.gov.in/compendium/Technology/2019%20%20Smart_sampling_odisha_NRSC_Tech%20Report.pdf, National Remote Sensing Centre (ISRO) and Department of Agriculture and Farmers Empowerment, Government of Odisha, Tech. Rep. NRSC-RSAA-ASAG-AGSciOff-March 2019-TR0001259-v1.0, Mar. 2019, accessed 2026-10-16; full PDF downloaded, front matter and abstract read.

[4] Mahalanobis National Crop Forecast Centre, Department of Agriculture and Farmers Welfare, Government of India, *Yield Estimation System based on Technology (YES-TECH) under PMFBY: Manual for Implementation*, https://apsac.ap.gov.in/wp-content/uploads/2023/01/Document2.pdf, Jan. 2023, accessed 2026-10-16; copy hosted by the Andhra Pradesh Space Applications Centre (state government); preface and sections 3.2 and 4.3.6 read.

[5] W. Kölle, M. Buchholz, and O. Musshoff, "Do high-resolution satellite indices at field level reduce basis risk of satellite-based weather index insurance?" *Agricultural Finance Review*, vol. 82, no. 4, p. 616–640, Aug. 2021. [Online]. Available: http://dx.doi.org/10.1108/afr-12-2020-0177

[6] T. Dalhaus and R. Finger, "Can gridded precipitation data and phenological observations reduce basis risk of weather index–based insurance?" *Weather, Climate, and Society*, vol. 8, no. 4, p. 409–419, Oct. 2016. [Online]. Available: http://dx.doi.org/10.1175/WCAS-D-16-0020.1

[7] M. Carter, A. de Janvry, E. Sadoulet, and A. Sarris, "Index insurance for developing country agriculture: A reassessment," *Annual Review of Resource Economics*, vol. 9, no. 1, p. 421–438, Oct. 2017. [Online]. Available: http://dx.doi.org/10.1146/annurev-resource-100516-053352

[8] F. Ceballos, B. Kramer, and M. Robles, "The feasibility of picture-based insurance (pbi): Smartphone pictures for affordable crop insurance," *Development Engineering*, vol. 4, p. 100042, 2019. [Online]. Available: http://dx.doi.org/10.1016/j.deveng.2019.100042

[9] S. Cole, X. Giné, J. Tobacman, P. Topalova, R. Townsend, and J. Vickery, "Barriers to household risk management: Evidence from india," *American Economic Journal: Applied Economics*, vol. 5, no. 1, p. 104–135, Jan. 2013. [Online]. Available: http://dx.doi.org/10.1257/app.5.1.104

[10] The Tribune (India), "Insurance claims: Technical panel to look into mismatch of area under cotton," https://www.tribuneindia.com/news/haryana/insurance-claims-technical-panel-to-look-into-mismatch-of-area-under-cotton-572034/amp, Dec. 2023, news report, not peer reviewed; accessed 2026-10-16.

[11] Gemini Team, Google, "Gemini: A family of highly capable multimodal models," 2023. [Online]. Available: https://arxiv.org/abs/2312.11805

[12] G. Kim, T. Hong, M. Yim, J. Nam, J. Park, J. Yim, W. Hwang, S. Yun, D. Han, and S. Park, *OCR-Free Document Understanding Transformer*. Springer Nature Switzerland, 2022, p. 498–517. [Online]. Available: http://dx.doi.org/10.1007/978-3-031-19815-1_29

[13] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. J. Bang, A. Madotto, and P. Fung, "Survey of hallucination in natural language generation," *ACM Computing Surveys*, vol. 55, no. 12, p. 1–38, Mar. 2023. [Online]. Available: http://dx.doi.org/10.1145/3571730

[14] C. Rudin, "Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead," *Nature Machine Intelligence*, vol. 1, no. 5, p. 206–215, May 2019. [Online]. Available: http://dx.doi.org/10.1038/s42256-019-0048-x

[15] I. D. Raji, A. Smart, R. N. White, M. Mitchell, T. Gebru, B. Hutchinson, J. Smith-Loud, D. Theron, and P. Barnes, "Closing the ai accountability gap: defining an end-to-end framework for internal algorithmic auditing," in *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*, ser. FAT* '20. ACM, Jan. 2020, p. 33–44. [Online]. Available: http://dx.doi.org/10.1145/3351095.3372873

[16] M. Drusch, U. Del Bello, S. Carlier, O. Colin, V. Fernandez, F. Gascon, B. Hoersch, C. Isola, P. Laberinti, P. Martimort, A. Meygret, F. Spoto, O. Sy, F. Marchese, and P. Bargellini, "Sentinel-2: Esa's optical high-resolution mission for gmes operational services," *Remote Sensing of Environment*, vol. 120, p. 25–36, May 2012. [Online]. Available: http://dx.doi.org/10.1016/j.rse.2011.11.026

[17] C. J. Tucker, "Red and photographic infrared linear combinations for monitoring vegetation," *Remote Sensing of Environment*, vol. 8, no. 2, p. 127–150, May 1979. [Online]. Available: http://dx.doi.org/10.1016/0034-4257(79)90013-0

[18] A. Ursu, V. Istrate, V. Jitariu, and I.-L. Lazăr, "Hail damage detection: Integrating sentinel-2 images with weather radar hail kinetic energy," *Remote Sensing*, vol. 17, no. 23, p. 3850, Nov. 2025. [Online]. Available: http://dx.doi.org/10.3390/rs17233850

[19] T. Ha, Y. Shen, H. Duddu, E. Johnson, and S. J. Shirtliffe, "Quantifying hail damage in crops using sentinel-2 imagery," *Remote Sensing*, vol. 14,

no. 4, p. 951, Feb. 2022. [Online]. Available: http://dx.doi.org/10.3390/rs14040951

[20] F. Bocchino, V. Belloni, R. Ravanelli, C. Zaccarini, M. Crespi, and R. Lindenbergh, "Crop flood damage assessment integrating sentinel-2 imagery and in situ data: the 2023 emilia-romagna case," *Remote Sensing Applications: Society and Environment*, vol. 41, p. 101852, Jan. 2026. [Online]. Available: http://dx.doi.org/10.1016/j.rsase.2025.101852

[21] U. Kanjir, N. Đurić, and T. Veljanovski, "Sentinel-2 based temporal detection of agricultural land use anomalies in support of common agricultural policy monitoring," *ISPRS International Journal of Geo-Information*, vol. 7, no. 10, p. 405, Oct. 2018. [Online]. Available: http://dx.doi.org/10.3390/ijgi7100405

[22] A. Chakraborty, P. Srikanth, C. S. Murthy, P. V. N. Rao, and S. Chowdhury, "Assessing lodging damage of jute crop due to super cyclone amphan using multi-temporal sentinel-1 and sentinel-2 data over parts of west bengal, india," *Environmental Monitoring and Assessment*, vol. 193, no. 8, Jul. 2021. [Online]. Available: http://dx.doi.org/10.1007/s10661-021-09220-w

[23] L. Baetens, C. Desjardins, and O. Hagolle, "Validation of copernicus sentinel-2 cloud masks obtained from maja, sen2cor, and fmask processors using reference cloud masks generated with a supervised active learning procedure," *Remote Sensing*, vol. 11, no. 4, p. 433, Feb. 2019. [Online]. Available: http://dx.doi.org/10.3390/rs11040433

[24] K. Raiyani, T. Gonçalves, L. Rato, P. Salgueiro, and J. R. Marques da Silva, "Sentinel-2 image scene classification: A comparison between sen2cor and a machine learning approach," *Remote Sensing*, vol. 13, no. 2, p. 300, Jan. 2021. [Online]. Available: http://dx.doi.org/10.3390/rs13020300

[25] C. Aybar, L. Ysuhuaylas, J. Loja, K. Gonzales, F. Herrera, L. Bautista, R. Yali, A. Flores, L. Diaz, N. Cuenca, W. Espinoza, F. Prudencio, V. Llactayo, D. Montero, M. Sudmanns, D. Tiede, G. Mateo-García, and L. Gómez-Chova, "Cloudsen12, a global dataset for semantic understanding of cloud and cloud shadow in sentinel-2," *Scientific Data*, vol. 9, no. 1, Dec. 2022. [Online]. Available: http://dx.doi.org/10.1038/s41597-022-01878-2

[26] D. Sani, S. Mahato, S. Saini, H. K. Agarwal, C. Chandra Devshali, S. Anand, G. Arora, and T. Jayaraman, "Sickle: A multi-sensor satellite imagery dataset annotated with multiple key cropping parameters," in *2024 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*. IEEE, Jan. 2024, p. 5983–5992. [Online]. Available: http://dx.doi.org/10.1109/wacv57701.2024.00589

[27] D. B. Lobell, G. Azzari, M. Burke, S. Gourlay, Z. Jin, T. Kilic, and S. Murray, "Eyes in the sky, boots on the ground: Assessing satellite- and ground-based approaches to crop yield measurement and analysis," *American Journal of Agricultural Economics*, vol. 102, no. 1, p. 202–219, Oct. 2019. [Online]. Available: http://dx.doi.org/10.1093/ajae/aaz051

[28] M. Burke and D. B. Lobell, "Satellite-based assessment of yield variation and its determinants in smallholder african systems," *Proceedings of the National Academy of Sciences*, vol. 114, no. 9, p. 2189–2194, Feb. 2017. [Online]. Available: http://dx.doi.org/10.1073/pnas.1616919114

[29] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive nlp tasks," 2020. [Online]. Available: https://arxiv.org/abs/2005.11401

[30] Y. Gao, Y. Xiong, X. Gao, K. Jia, J. Pan, Y. Bi, Y. Dai, J. Sun, M. Wang, and H. Wang, "Retrieval-augmented generation for large language models: A survey," 2023. [Online]. Available: https://arxiv.org/abs/2312.10997

[31] D. J. S. Ravindran, I. Skarga-Bandurova, S. V, M. Awais, and M. S, "Agrollm: Connecting farmers and agricultural practices through large language models for enhanced knowledge transfer and practical application," *AgriEngineering*, vol. 8, no. 1, p. 38, Jan. 2026. [Online]. Available: http://dx.doi.org/10.3390/agriengineering8010038

[32] A. Balaguer, V. Benara, R. L. d. F. Cunha, R. d. M. E. Filho, T. Hendry, D. Holstein, J. Marsman, N. Mecklenburg, S. Malvar, L. O. Nunes, R. Padilha, M. Sharp, B. Silva, S. Sharma, V. Aski, and R. Chandra, "Rag vs fine-tuning: Pipelines, tradeoffs, and a case study on agriculture," 2024. [Online]. Available: https://arxiv.org/abs/2401.08406

[33] N. Singh, J. Wang'ombe, N. Okanga, T. Zelenska, J. Repishti, J. G. K, S. Mishra, R. Manokaran, V. Singh, M. I. Rafiq, R. Gandhi, and A. Nambi, "Farmer.chat: Scaling ai-powered agricultural services for smallholder farmers," 2024. [Online]. Available: https://arxiv.org/abs/2409.08916

[34] J. Kpodo, P. Kordjamshidi, and A. P. Nejadhashemi, "Agxqa: A benchmark for advanced agricultural extension question answering," *Computers and Electronics in Agriculture*, vol. 225, p. 109349, Oct. 2024. [Online]. Available: http://dx.doi.org/10.1016/j.compag.2024.109349

[35] M. Z. U. Rehman, D. Raghuvanshi, and N. Kumar, "Kisanqrs: A deep learning-based automated query-response system for agricultural decision-making," *Computers and Electronics in Agriculture*, vol. 213, p. 108180, Oct. 2023. [Online]. Available: http://dx.doi.org/10.1016/j.compag.2023.108180

[36] K. Didwania, P. Seth, A. Kasliwal, and A. Agarwal, "Agrillm:harnessing transformers for framer queries," in *Proceedings of the Third Workshop on NLP for Positive Impact*. Association for Computational Linguistics, 2024, p. 179–187. [Online]. Available: http://dx.doi.org/10.18653/v1/2024.nlp4pi-1.16

[37] H. Joren, J. Zhang, C.-S. Ferng, D.-C. Juan, A. Taly, and C. Rashtchian, "Sufficient context: A new lens on retrieval augmented generation systems," 2024, published at ICLR 2025; proceedings page https://proceedings.iclr.cc/paper_files/paper/2025/hash/33dffa2e3d2ab74a783d1a8c292f66d9-Abstract-Conference.html. [Online]. Available: https://arxiv.org/abs/2411.06037

[38] S. Es, J. James, L. Espinosa Anke, and S. Schockaert, "Ragas: Automated evaluation of retrieval augmented generation," in *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*. Association for Computational Linguistics, 2024, p. 150–158. [Online]. Available: http://dx.doi.org/10.18653/v1/2024.eacl-demo.16

[39] S. A. Cole and A. N. Fernando, "'mobile'izing agricultural advice technology adoption diffusion and sustainability," *The Economic Journal*, vol. 131, no. 633, p. 192–219, Jul. 2020. [Online]. Available: http://dx.doi.org/10.1093/ej/ueaa084

[40] R. Shinoda, N. Inoue, H. Kataoka, M. Onishi, and Y. Ushiku, "Agrobench: Vision-language model benchmark in agriculture," in *2025 IEEE/CVF International Conference on Computer Vision (ICCV)*. IEEE, Oct. 2025, p. 7634–7644. [Online]. Available: http://dx.doi.org/10.1109/iccv51701.2025.00716

[41] "Statistical API – copernicus data space ecosystem documentation," https://documentation.dataspace.copernicus.eu/APIs/SentinelHub/Statistical.html, accessed 2026-10-16; only page title and section headings were read.

[42] Open-Meteo, "Historical weather API," https://open-meteo.com/en/docs/historical-weather-api, accessed 2026-10-16; page states ERA5 (0.25 deg, from 1940) and ERA5-Land (0.1 deg, from 1950) reanalysis.

AUTHOR BIOGRAPHIES WITHHELD IN THIS DRAFT IEEE Access requires a short biography of every author, placed below the references. The biographies will be added with the author names before submission.